

This section introduces a new class of zero-knowledge protocols which are based on the Sum-Check protocol. If previously we mostly worked over univariate polynomials, Sum-Check derivatives typically operate over multilinear polynomials in many variables. Turns out that such representation can give a competitive performance to protocols relying on high-degree univariate polynomials.

Besides Sum-Check-based approaches, we also introduce the notion of *lookup checks*, which is a quite recent development, allowing to significantly reduce the arithmetical circuit size for non-native problems for the a zero-knowledge protocol (such as hash function computation or zero-knowledge Machine Learning). While this is a standalone research direction, we still provide it after discussing Sum-Check protocol since historically one of the first lookup protocols (namely, logup) is in fact Sum-Check-based.

Specifically, we list sections of this chapter in ??.

Section	Topic	Key Concepts
??	Sum-Check Protocol	Construction of Sum-Check protocol, its performance and soundness
??	GKR Protocol	Sum-Check-based SNARK over arithmetical circuits
??	Lookup Checks	Motivation for lookup checks; plookup and logup constructions
??	UltraGroth	Implementing lookup checks in Groth16

Table 1: Topics covered in Part IV